

Beyond the Wronskian

Investigating Necessary vs. Sufficient Conditions
for Linear Independence of ODE Solutions

MATHEMATICS IV PROJECT · 2025-26

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1 Introduction

When I first studied linear ordinary differential equations in Mathematics IV, I learned that the Wronskian determinant is a practical tool to test whether two functions are linearly independent. The standard statement appears in both of our main references: if the Wronskian is nonzero at even a single point on an interval, the functions are linearly independent (Kreyszig, Sec. 2.8, p. 91; Ross, Sec. 4.1, Theorem 4.4, p. 112). This is a **sufficient condition**: a non-zero Wronskian guarantees independence.

Naturally, I asked myself the reverse question: if the Wronskian is zero everywhere on an interval, does it automatically mean the functions are linearly dependent? At first glance, many of the examples I solved in class seemed to suggest the answer was “yes.” However, while reading Kreyszig more carefully, I came across a remark that surprised me:

Kreyszig §2.8, Example 2 (p. 92) — Continuity assumption cannot be omitted

“...if u_1 and u_2 are any differentiable functions of x , the condition $W(u_1, u_2) = 0$ is only necessary for linear dependence of u_1 and u_2 [as follows from part (a) of the proof of Theorem 2], but not sufficient.”

The concrete pair Kreyszig gives is $f_1(x) = x^2$ and $f_2(x) = x|x|$ on $[-1, 1]$: their Wronskian vanishes identically, yet they are clearly linearly independent. Understanding *why* this does not contradict our ODE theory was the starting point for this project.

The resolution turns out to hinge on a beautiful result called **Abel's Identity**. For genuine solutions of a second-order linear ODE with continuous coefficients, the Wronskian satisfies a first-order ODE itself, forcing it to be either always zero or never zero on the interval. That all-or-nothing behaviour is what makes the Wronskian a necessary *and* sufficient test for ODE solutions — even though it is merely sufficient for arbitrary functions.

In what follows I will: (i) recall the precise definitions; (ii) prove the sufficient condition carefully; (iii) work through the counterexample with full calculation and figures; (iv) state and explain Abel's Identity; and (v) illustrate the well-behaved ODE case with $\sin x$ and $\cos x$. All figures were produced by me using R.

2 Mathematical Preliminaries

The following definitions follow Ross §4.1, pp. 105–111 and Kreyszig §2.8, pp. 90–91 exactly as presented in our course.

2.1 Linear Dependence and Independence

Definition 2.1 — Linear Dependence

A set of functions $f_1(x)$, $f_2(x)$, ..., $f_n(x)$ on an interval I is linearly dependent on I if there exist constants c_1 , c_2 , ..., c_n , not all zero, such that

$$c_1 f_1(x) + c_2 f_2(x) + \dots + c_n f_n(x) = 0 \quad \text{for all } x \text{ in } I.$$

If only the trivial combination works, the functions are linearly independent.

For two functions, linear dependence means simply that one is a constant multiple of the other on the *entire* interval — it is not enough for them to agree only at a single point (Ross, p. 105). This seemingly simple point becomes crucial when we examine the counterexample in Section 4.

2.2 The Wronskian

Definition 2.2 (Wronskian of Two Functions)

Let $f_1(x)$ and $f_2(x)$ be differentiable on I . Their Wronskian is the determinant

$$W[f_1, f_2](x) = \begin{vmatrix} f_1(x) & f_2(x) \\ f_1'(x) & f_2'(x) \end{vmatrix} = f_1(x) f_2'(x) - f_1'(x) f_2(x).$$

Definition 2.3 (Wronskian of n Functions)

If $f_1, \dots, f_n$ have at least $n-1$ derivatives, the Wronskian is the $n \times n$ determinant:

$$W[f_1, \dots, f_n](x) = \begin{vmatrix} f_1 & \cdots & f_n \\ f_1' & \cdots & f_n' \\ \vdots & \ddots & \vdots \\ f_1^{(n-1)} & \cdots & f_n^{(n-1)} \end{vmatrix}$$

The Wronskian is itself a function of x . Whether it is zero or nonzero at a particular point — or everywhere on I — is the crux of everything that follows.

2.3 A Key Property for ODE Solutions

When the functions happen to be solutions of a linear homogeneous ODE with continuous coefficients, the Wronskian satisfies Abel's Identity (Kreyszig, p. 91; Ross, Theorem 4.5, p. 112). This identity will be the key to understanding why the Wronskian works so cleanly in ODE theory, as I will discuss in Section 5.

3 The Sufficient Condition: $W \neq 0 \Rightarrow$ Independence

Theorem 3.1 — Sufficient Condition (Ross Thm 4.4, p. 112; Kreyszig §2.8, p. 90)

If f_1 and f_2 are differentiable on I and there exists x_0 in I with $W[f_1, f_2](x_0) \neq 0$, then f_1 and f_2 are linearly independent on I .

3.1 Proof by Contrapositive

We prove the logically equivalent contrapositive: if f_1 and f_2 are linearly dependent on I , then $W(x) = 0$ for all $x \in I$.

Proof. Suppose f_1 and f_2 are linearly dependent. Then there exist constants c_1, c_2 , not both zero, with

$$c_1 f_1(x) + c_2 f_2(x) = 0 \quad \text{for all } x \in I.$$

Differentiating:

$$c_1 f_1'(x) + c_2 f_2'(x) = 0 \quad \text{for all } x \in I.$$

For each fixed x these two equations form a homogeneous linear system in (c_1, c_2) . Since a non-trivial solution exists, the coefficient determinant must be zero:

$$f_1(x) f_2'(x) - f_1'(x) f_2(x) = W[f_1, f_2](x) = 0.$$

This holds for every $x \in I$. ■

3.2 Example

For a quick illustration, the functions $\sin x$ and $\cos x$ have Wronskian

$$W[\sin x, \cos x](x) = \sin x \cdot (-\sin x) - \cos x \cdot \cos x = -1 \neq 0.$$

which is never zero. Therefore, by Theorem 3.1, they are linearly independent on any interval (Ross, Example 4.14, p. 112; Kreyszig, Example 1, §2.8, p. 91).

3.3 The Natural Question

Since a non-zero Wronskian is sufficient for independence, it is natural to ask whether it is also necessary. That is, if $W(x) = 0$ everywhere, must the functions be dependent? In the next section I will show that the answer is, surprisingly, no.

4 The Counterexample: $W = 0$ Does Not Imply Dependence

I was genuinely surprised when I first encountered a counterexample that disproves the converse. It comes directly from Kreyszig (Example 2, p. 92, and Problem 20, p. 93).

4.1 The Functions $f_1(x) = x^2$ and $f_2(x) = x|x|$

On the interval $I = [-1, 1]$ define:

$$f_1(x) = x^2, \quad f_2(x) = x|x|$$

where $f_2(x) = x^2$ for $x \geq 0$, and $f_2(x) = -x^2$ for $x < 0$.

Both functions are differentiable on I . The derivative of f_2 is $f_2'(x) = 2|x|$, which is continuous everywhere (including at $x = 0$ where both one-sided limits equal 0). Notice that on the positive half-line the two functions *coincide*: both equal x^2 . On the negative half-line they differ in sign. This asymmetry is what makes the example work.

4.2 Proving Linear Independence

Suppose c_1 and c_2 are constants with $c_1 x^2 + c_2 x|x| = 0$ for all $x \in [-1, 1]$.

For $x > 0$: $f_2(x) = x^2$, so the equation becomes $(c_1 + c_2) x^2 = 0$, giving $c_1 + c_2 = 0$.

For $x < 0$: $f_2(x) = -x^2$, so the equation becomes $(c_1 - c_2) x^2 = 0$, giving $c_1 - c_2 = 0$.

Adding and subtracting: $2c_1 = 0$ and $2c_2 = 0$, so $c_1 = c_2 = 0$. The only solution is the trivial one, confirming **f_1 and f_2 are linearly independent on $[-1, 1]$** .

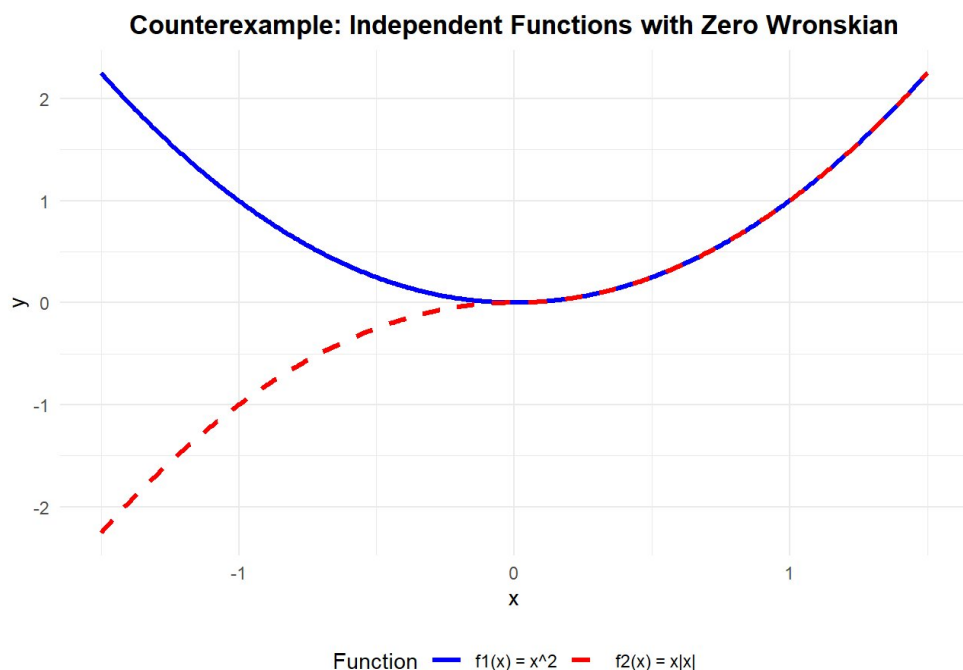


Figure 1. The functions $f_1(x) = x^2$ (blue, solid) and $f_2(x) = x|x|$ (red, dashed). They overlap for $x \geq 0$ but diverge for $x < 0$ — no single constant multiple can make them equal on the whole interval.

4.3 The Vanishing Wronskian

I compute $W[f_1, f_2](x)$ piecewise. Recall $f_1'(x) = 2x$ throughout.

Case $x \geq 0$: $f_2 = x^2$, $f_2' = 2x$.

$$W(x) = x^2 \cdot 2x - 2x \cdot x^2 = 2x^3 - 2x^3 = 0.$$

Case $x < 0$: $f_2 = -x^2$, $f_2' = -2x$.

$$W(x) = x^2 \cdot (-2x) - 2x \cdot (-x^2) = -2x^3 + 2x^3 = 0.$$

Continuity of W at $x = 0$ is clear since each branch approaches 0. Therefore **$W[f_1, f_2](x) \equiv 0$ on $[-1, 1]$** , confirmed by Figure 2.

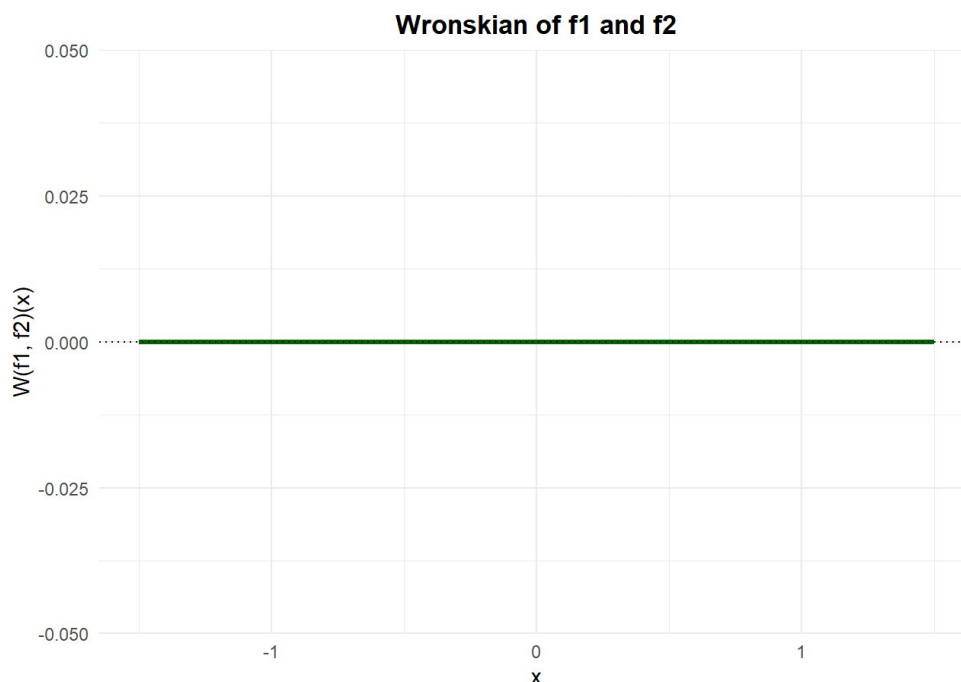


Figure 2. The Wronskian $W[f_1, f_2](x)$ is identically zero on $[-1, 1]$, even though f_1 and f_2 are linearly independent.

4.4 Conclusion: A False Necessary Condition

This example establishes that for *arbitrary differentiable functions*:

Statement	True for arbitrary differentiable functions?
$W \neq 0$ at some point $\Rightarrow$ independent	YES (Theorem 3.1)
Dependent $\Rightarrow W \equiv 0$	YES (proof in §3.1)
$W \equiv 0 \Rightarrow$ dependent	NO (counterexample above)

So $W = 0$ is a **necessary** consequence of dependence but is **not sufficient** to guarantee dependence. This is exactly the subtlety Kreyszig flags.

But wait — in every ODE problem we do in class, $W \equiv 0$ *does* mean the solutions are dependent. Why? The answer lies in the special structure that ODE solutions must obey, explored in the next section.

5 The Resolution: ODE Solutions and Abel's Identity

The counterexample functions x^2 and $x|x|$ are solutions of

$$y'' - \frac{3}{x}y' + \frac{3}{x^2}y = 0,$$

as one can verify by substitution. But notice carefully: they *do* satisfy an ODE — just not one with **continuous** coefficients. The coefficient $p(x) = -3/x$ blows up at $x = 0$, violating the continuity hypothesis of Kreyszig's Theorem 2 (§2.8). This is the critical distinction: these functions satisfy *some* ODE, but not under the hypotheses required for Abel's Identity to apply. When the coefficients *are* continuous, the loophole closes completely.

5.1 Abel's Identity

Abel's Identity (Kreyszig §2.8, p. 91; Ross Theorem 4.5, p. 112)

Let y_1, y_2 be any two solutions of $y'' + p(x)y' + q(x)y = 0$
where p and q are continuous on an open interval I .
Then for any x_0 in I :

$$W(x) = W(x_0) \cdot \exp\left(-\int_{x_0}^x p(t) dt\right).$$

$$W(x) = W(x_0) \exp\left(-\int_{x_0}^x p(t) dt\right).$$

5.2 The Equivalence Theorem

Since the exponential factor is *never* zero, the sign of $W(x)$ is determined entirely by $W(x_0)$. Therefore:

If ...	Then W is ...	Consequence
$W(x_0) \neq 0$ for some x_0	never zero on all of I	solutions are independent
$W(x_0) = 0$ for some x_0	identically 0 on all of I	solutions are dependent

This is the **all-or-nothing property** of the Wronskian for ODE solutions. It immediately upgrades the Wronskian test from merely sufficient to necessary-and-sufficient:

Equivalence Theorem (Kreyszig Thm 2, §2.8, p. 91; Ross Thms 4.4–4.5, p. 112)

Under the continuity hypothesis on p and q :

y_1, y_2 linearly dependent $\Leftrightarrow W[y_1, y_2] = 0$ at some (and thus all) points of I .
 y_1, y_2 linearly independent $\Leftrightarrow W[y_1, y_2] \neq 0$ everywhere on I .

The key contrast with Section 4 is the word '*solutions*'. Arbitrary differentiable functions are not constrained by any ODE, so Abel's Identity has no reason to hold for them — and indeed it does not, as our counterexample shows.

5.3 Sketch of Why Abel's Identity Holds

Consider the ODE $y'' + p(x)y' + q(x)y = 0$ with solutions y_1 and y_2 . Differentiating the Wronskian $W = y_1 y_2' - y_1' y_2$ gives:

$$W' = (y_1' y_2' + y_1 y_2'') - (y_1'' y_2 + y_1' y_2') = y_1 y_2'' - y_1'' y_2.$$

The $y_1' y_2'$ terms cancel. Since each y_i solves the ODE, $y_i'' = -p y_i' - q y_i$. Substituting:

$$W' = y_1(-p y_2' - q y_2) - (-p y_1' - q y_1)y_2 = -p(y_1 y_2' - y_1' y_2) = -p W.$$

So W satisfies the separable ODE $W' = -p(x)W$, which has the unique solution:

$$W(x) = W(x_0) e^{-\int_{x_0}^x p(t) dt}.$$

This is Abel's Identity. The exponential is never zero, so W is either identically zero (if $W(x_0) = 0$) or never zero (if $W(x_0) \neq 0$). ■

5.4 Illustration: $\sin x$ and $\cos x$ for $y'' + y = 0$

Here $p(x) = 0$, so the formula gives $W(x) = W(x_0) \cdot e^0 = W(x_0)$. Indeed the Wronskian is constant:

$$W[\sin x, \cos x](x) = -\sin^2 x - \cos^2 x = -1 \quad (\text{constant, nonzero}).$$

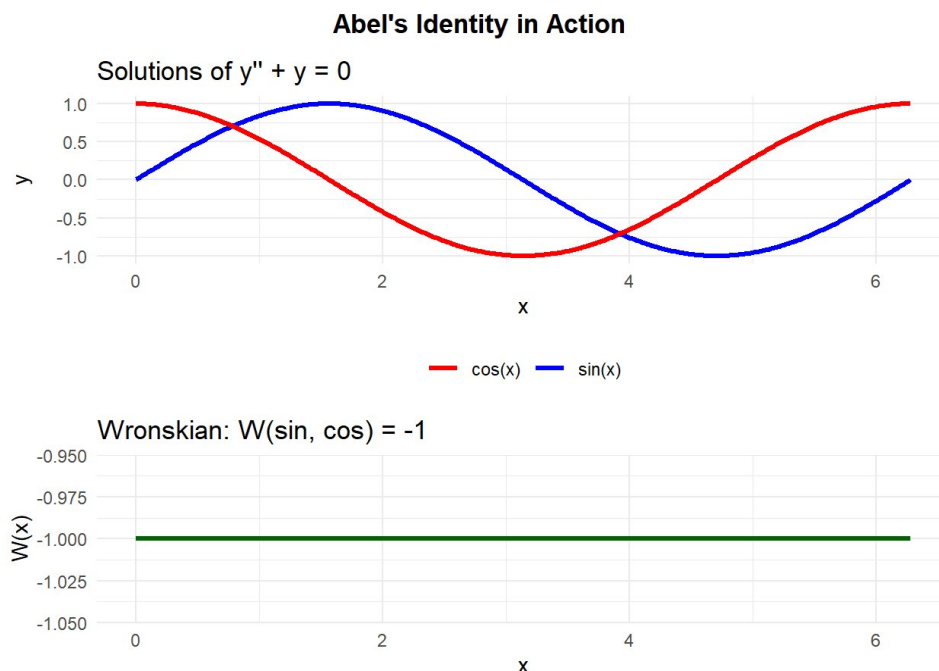


Figure 3. (top) Solutions $\sin x$ (blue) and $\cos x$ (red) of $y'' + y = 0$. (bottom) Their Wronskian is the constant -1 , consistent with Abel's Identity ($p = 0$ here). Since W is never zero the solutions are linearly independent on every interval.

This is a completely clean illustration of the all-or-nothing property: the Wronskian does not even change with x , let alone vanish.

5.5 Implication for Fundamental Sets

Because the Wronskian of independent ODE solutions is nonzero everywhere, any set of n independent solutions of an n -th order linear ODE forms a **fundamental set of solutions** (basis). This guarantees that the general solution $c_1 y_1 + c_2 y_2 + \dots + c_n y_n$ captures every solution of the ODE — nothing is left out. That is the foundational justification for the solution method we use throughout the course (Ross, p. 111; Kreyszig, p. 92).

6 Conclusion

This project started from a simple question: does $W \equiv 0$ force linear dependence? The answer I found is:

Condition	Arbitrary differentiable functions	ODE solutions: $y''+p(x)y'+q(x)y=0$ (p, q continuous)
$W \neq 0$ at some point $\Rightarrow$ independent	YES (sufficient)	YES (sufficient)
Dependent $\Rightarrow W \equiv 0$	YES (necessary)	YES (necessary)
$W \equiv 0 \Rightarrow$ dependent	NO — counterexample: x^2 and $x x$	YES (Abel's Identity)

Three things I take away from working through this project:

- **The counterexample is instructive, not pathological.** The functions x^2 and $x|x|$ are perfectly well-behaved and natural. What makes them special is that they are *not* solutions of any ODE with continuous coefficients — and that is precisely what the counterexample is telling us.
- **Abel's Identity is the key theorem.** It is the reason the Wronskian works so cleanly in ODE theory. Without it, the Wronskian test would be a one-way street, as it is for arbitrary functions.
- **Hypotheses matter.** A theorem stated for ODE solutions cannot be applied casually to any pair of functions. The continuity of p and q is not a cosmetic assumption — removing it invalidates the conclusion. This is a lesson I'll carry into every subsequent theorem in the course.

7 References

- [1] **Kreyszig, E.** (2011). *Advanced Engineering Mathematics* (10th ed.). Hoboken, NJ: John Wiley & Sons. — Sec. 2.8: Existence and Uniqueness of Solutions, pp. 87–93; Sec. 2.9: Homogeneous Linear Equations of Arbitrary Order, pp. 94–100.
- [2] **Ross, S. L.** (1984). *Differential Equations* (3rd ed.). New York, NY: John Wiley & Sons. — Sec. 4.1: Basic Theory of Linear Differential Equations, pp. 102–124.